IT4010 Research Project – 2025 July Topic Assessment Form

Project ID : 25-26J-034

1. Topic (12 words max)

AI and IoT-Enabled Smart Office System with Cloud-Based Energy Optimization

2. Research group the project belongs to

AIMS - Autonomous Intelligent Machines and Systems

3. Specialization of the project belongs to

Computer Systems and Network Engineering(CSNE)

4. If a continuation of a previous project:

Project ID	
Year	

5. Brief description of the research problem including references (200 – 500 words max) – references not included in word count.

The growing demand for energy efficiency in modern infrastructure has made smart building technologies an essential area of research and development. Buildings account for nearly 40% of global energy consumption, with a significant portion wasted due to inefficient monitoring and control systems (International Energy Agency, 2021). Traditional energy management systems often rely on manual readings, fixed schedules, or isolated sensors, which lack the ability to provide holistic insights into real-time energy consumption. As a result, abnormal usage patterns frequently go undetected, leading to unnecessary energy costs, equipment strain, and reduced sustainability.

The emergence of the Internet of Things (IoT) has created opportunities to address these challenges by enabling real-time energy data collection through sensor-enabled devices. However, IoT systems alone face limitations such as high data volume, reliability issues, and difficulties in extracting actionable insights. Similarly, cloud platforms have become popular for large-scale storage and analytics, but without proper integration with IoT-based data sources, their potential remains underutilized (Alahakoon & Yu, 2016). The gap lies in creating an integrated solution that can combine IoT sensing, wireless communication, and cloud analytics to deliver both **real-time visibility** and **intelligent alerting** for energy optimization.

Smart buildings require more than just consumption monitoring; they need intelligent systems that can detect anomalies, predict inefficiencies, and support proactive decision-making. For instance, sudden spikes in current or abnormal power patterns may indicate faulty equipment or misused devices. Without an automated alerting mechanism, such issues remain unnoticed until significant damage or energy waste occurs. Existing solutions are often fragmented—some focus on IoT sensing, others on cloud dashboards—yet very few provide an end-to-end approach integrating data acquisition, transmission, analysis, visualization, and alerting.

This research problem centers on developing an IoT-enabled, cloud-integrated energy monitoring system tailored for smart buildings. By leveraging IoT devices for real-time data acquisition, cloud platforms for scalable analytics, and AI-driven rules for abnormal pattern detection, the system aims to enhance energy efficiency, reduce operational costs, and improve sustainability outcomes. Addressing this problem contributes not only to smart building optimization but also to broader sustainability goals such as reducing greenhouse gas emissions and supporting global energy conservation efforts.

References:

Assessing energy saving potentials of office buildings based on adaptive thermal comfort using a tracking-based method (2019) Ming, R., Yu, W., Zhao, X., Liu, Y., Li, B., Essah, E. ORCID International Energy Agency (IEA). (2021). *Energy Efficiency 2021*. IEA.Alahakoon, D., & Yu, X. (2016). Smart Electricity Meter Data Intelligence for Future Energy Systems: A Survey. *IEEE Transactions on Industrial Informatics*, 12(1), 425–436.

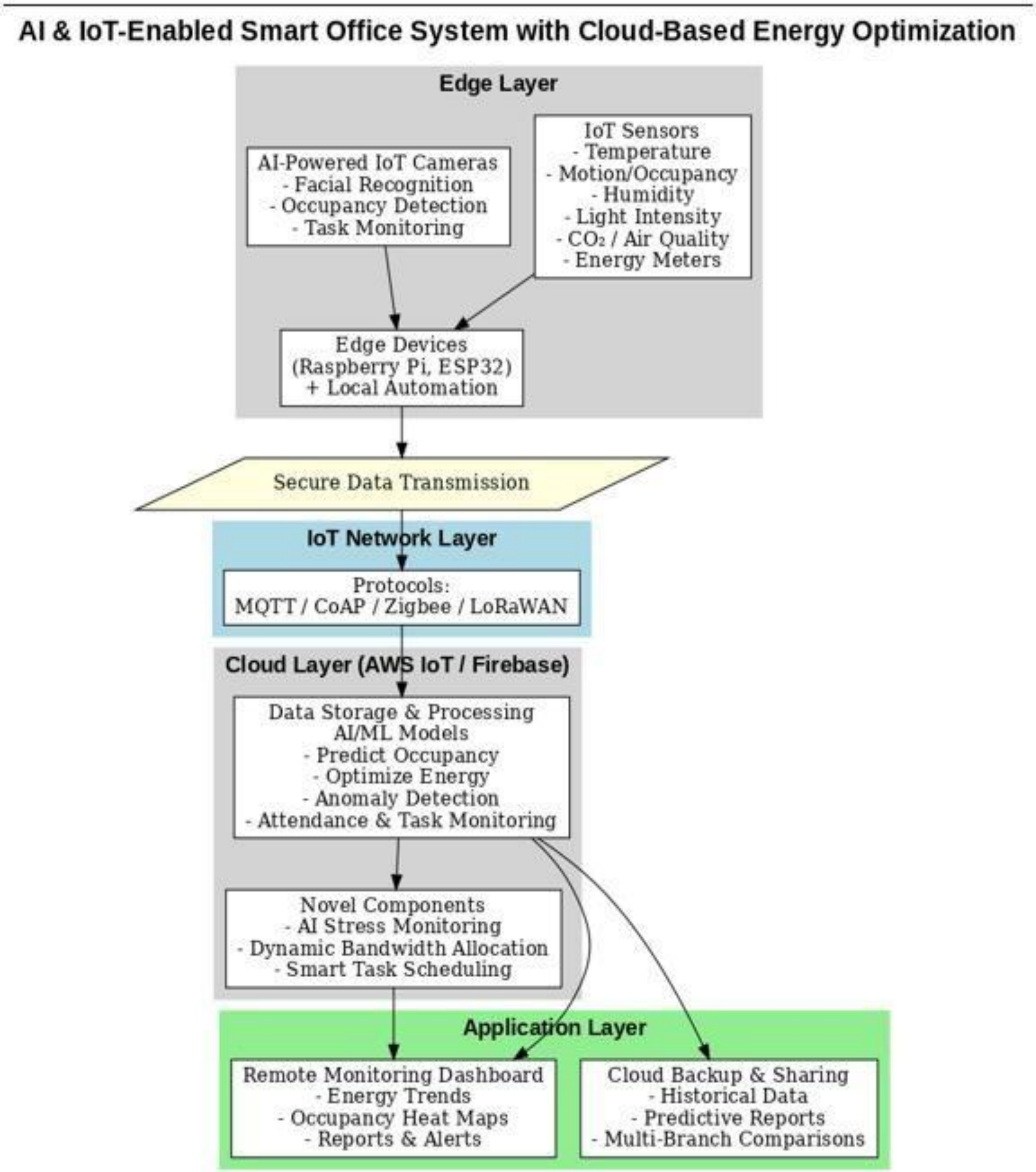
6. Brief description of the nature of the solution including a conceptual diagram (250 words max)

The proposed solution is a **unified smart office automation framework** that integrates **AI-powered IoT cameras, IoT sensors, device automation, and cloud-based analytics** into a single system. The goal is to improve energy efficiency, streamline employee management, and provide real-time visibility into office operations.

At the **edge layer**, IoT-enabled cameras and sensors capture real-time data, such as occupancy, employee presence, and environmental parameters. AI models embedded on these devices enable **face recognition for attendance**, **activity detection for task monitoring**, and **context-aware automation** of devices such as lights, air conditioning, and appliances. For example, when no employees are detected, the system automatically powers down non-essential devices to save energy.

At the **cloud layer**, data collected from edge devices is transmitted securely to platforms such as **AWS IoT Core or Firebase**. Here, scalable storage and **predictive analytics** are applied to optimize energy usage patterns, monitor environmental quality, and provide dashboards for administrators. The cloud also hosts the **rule-based alert system**, notifying managers of anomalies such as excessive energy consumption or unauthorized activity.

The integration of **edge intelligence with cloud analytics** ensures both **real-time responsiveness** and **long-term optimization**. By combining automation, attendance tracking, task scheduling, and predictive energy management into one framework, the solution addresses gaps in existing smart office systems and provides a scalable model for future deployment.


AI & IoT-Enabled Smart Office System with Cloud-Based Energy Optimization
Edge Layer
AI-Powered IoT Cameras
- Facial Recognition
- Occupancy Detection
- Task Monitoring
IoT Sensors
- Temperature
- Motion/Occupancy
- Humidity
- Light Intensity
- CO2 / Air Quality
- Energy Meters
Edge Devices
(Raspberry Pi, ESP32)
+ Local Automation
Secure Data Transmission
IoT Network Layer
Protocols:
MQTT / CoAP / Zigbee / LoRaWAN
Cloud Layer (AWS IoT / Firebase)
Data Storage & Processing
AI/ML Models
- Predict Occupancy
- Optimize Energy
- Anomaly Detection
- Attendance & Task Monitoring
Novel Components
- AI Stress Monitoring
- Dynamic Bandwidth Allocation
- Smart Task Scheduling
Application Layer
Remote Monitoring Dashboard
- Energy Trends
- Occupancy Heat Maps
- Reports & Alerts
Cloud Backup & Sharing
- Historical Data
- Predictive Reports
- Multi-Branch Comparisons

7. Brief description of specialized domain expertise, knowledge, and data requirements (300 words ma

The project requires specialized expertise across **IoT systems, artificial intelligence, energy management, and cloud computing**. Knowledge of **IoT device integration** and **sensor networks** is essential to monitor office parameters such as temperature, lighting, occupancy, and energy consumption. Proficiency in **embedded systems** and **wireless communication protocols** (e.g., Wi-Fi, Zigbee, or LoRaWAN) ensures reliable real-time data acquisition from multiple smart devices.

In the AI domain, expertise in **machine learning and predictive analytics** is needed to analyze sensor data, predict occupancy patterns, and optimize energy usage. Familiarity with **reinforcement learning** or **rule-based control algorithms** can enhance automated decision-making for dynamic control of HVAC, lighting, and other office systems. Knowledge of **data preprocessing, feature engineering, and model deployment** is important for implementing efficient AI models on either edge devices or cloud platforms.

Cloud computing expertise is required for **centralized data storage, analytics, and remote monitoring**, including experience with **cloud databases, IoT platforms, and real-time dashboards**. Integration with **energy management systems** and the ability to implement **scalable cloud architectures** are necessary to handle continuous streams of sensor data while maintaining low latency and secure access.

The data requirements include **continuous sensor readings** (temperature, occupancy, light, energy consumption), **historical energy usage records**, and **user behavior patterns** for AI training. Both **real-time streaming data** for immediate control and **aggregated historical data** for long-term optimization are needed. Additionally, metadata on device performance and energy efficiency metrics are essential to refine the AI models and validate energy optimization strategies.

8. Objectives and Novelty

Main Objective			
To design, develop, and implement a comprehensive AI-driven smart office automation ecosystem that seamlessly integrates IoT-enabled sensors, intelligent cameras, and cloud-based analytics to achieve optimal energy efficiency, automated employee management, and real-time operational intelligence. The system will leverage machine learning algorithms for predictive energy optimization, dynamic resource allocation, and proactive workplace wellness management, while providing scalable, secure, and sustainable solutions that reduce operational costs by up to 30% and enhance overall productivity in modern smart building environments.			
Member Name with Registration No	Sub Objective	Tasks	Novelty

M.A.H.S.Thathsarani IT22564672	-Ensure accurate and reliable sensor readings. -Establish stable IoT device connectivity. -Collect and preprocess data for AI analysis. -Enable basic local automation (lights, HVAC). -Maintain system reliability and temporary data logs. -Design for easy scalability with additional devices.	AI-Based Employee Current Report Generator System using Camera This component leverages IoT-enabled cameras and AI algorithms to automatically monitor employee activities and generate real-time reports. The system captures key metrics such as attendance, task engagement, and movement patterns, converting them into actionable insights. By automating report generation, it reduces manual effort and ensures	The AI-Based Smart Live Employee Stress Management System is novel because it combines real-time monitoring, AI analysis, and IoT devices to detect employee stress continuously. Unlike traditional surveys or periodic check-ins, it enables instant stress detection and proactive interventions, improving both workplace productivity and employee wellbeing. Additionally, it provides data-driven insights to management, allowing informed decisions for better workload and wellness management.

		accurate, timely information for management. AI-Based Smart Live Employee Stress Management System The stress management module uses AI and machine learning models to analyze live data collected from sensors and cameras. It detects signs of employee stress, fatigue, or discomfort in real-time, enabling proactive interventions. This smart system not only enhances workplace wellness but also helps organizations maintain productivity and employee satisfaction. Smart Task Scheduling & Monitoring System This module combines IoT data with AI-driven task management to optimize scheduling and monitor progress. Tasks are assigned intelligently based on	

		employee workload, availability, and performance metrics, while real-time monitoring ensures deadlines are met efficiently. This system streamlines operations, enhances accountability, and supports data-driven decision-making for project management.	
Shaethayine R. *(Leader)* IT22315700	-Set up cloud infrastructure for data storage and processing. -Manage real-time data ingestion and preprocessing for AI. -Develop AI models to predict occupancy and optimize energy usage. -Enable real-time automated control of office systems via AI. -Create dashboards for monitoring	Workload Data Collection The system begins by monitoring employee activities and tasks using IoT devices and software logs. Real-time data on task type, application usage, and activity levels are collected to evaluate each employee's workload accurately. Bandwidth Categorization and AI Analysis Collected workload data is processed using AI and machine learning models to categorize employees into Low, Medium, or High workload groups. Based on	The Bandwidth Categorization and AI Analysis component is novel because it uses AI and machine learning to analyze employee workload in real-time and dynamically allocate network bandwidth accordingly. Unlike traditional static allocation systems, this approach ensures that high-priority tasks receive sufficient resources while optimizing overall network efficiency. By combining workload intelligence with automated bandwidth management, the system enhances both employee productivity and organizational resource utilization, offering a proactive and adaptive solution not commonly found in existing workplace network management tools.

	energy consumption and automation status. -Test and optimize AI algorithms for accuracy and efficiency.	these categories, the system determines the appropriate bandwidth (Mbps) needed for each employee to perform tasks efficiently. Dynamic Bandwidth Allocation The network management system, using SD-WAN or QoS policies, dynamically allocates bandwidth according to the AI-determined categories. This ensures that high-priority tasks receive sufficient network resources while optimizing overall bandwidth usage across all employees.	

8. Individual component description of how it is complied with the specialization.

Member Name with Registration No	Description
M.A.H.S.Thathsarani IT22564672	The project incorporates **IoT sensors and edge devices** to monitor environmental and energy parameters in the office. This includes sensor selection, network configuration, data acquisition, and local automation. The work demonstrates expertise in **embedded programming, system reliability, and secure communication** across multiple devices, using protocols such as Wi-Fi, Zigbee, or LoRaWAN. These tasks align closely with the Computer System and Network Engineering specialization by combining hardware, software, and networking principles to ensure accurate, real-time data collection for the smart office environment.
Shaethayine R. *(Leader)* IT22315700	The project also involves **cloud infrastructure setup and AI-driven energy optimization**, including real-time data ingestion, predictive AI model development, and visualization dashboards for monitoring and automation. This component applies **distributed computing, networking for reliable data transfer, and end-to-end system integration**, connecting IoT devices with cloud-based analytics. It demonstrates the application of computer system and network engineering expertise to automate decision-making, optimize energy usage, and provide an intelligent, responsive smart office solution.

9. Supervisor details

	Title	First Name	Last Name	Signature
Supervisor				
Co-Supervisor				
External Supervisor				
Summary of external supervisor's (if any) experience and expertise				

This part is to be filled by the Topic Screening Staff members.

a) Does the chosen research topic possess a comprehensive scope suitable for a final-year project?

Yes		No	

b) Does the proposed topic exhibit novelty?

Yes		No	

c) Do you believe they have the capability to successfully execute the proposed project?

Yes		No	

d) Do the proposed sub-objectives reflect the students' areas of specialization?

Yes		No	

e) Supervisor's Evaluation and Recommendation for the Research topic:

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Acceptable: Mark/Select as necessary

Topic Assessment Accepted	
Topic Assessment Accepted with minor changes*	
Topic Assessment to be Resubmitted with major changes*	
Topic Assessment Rejected. Topic must be changed	

* Detailed comments given below

Comments

Staff Member's Name	Signature

*Important:

1. According to the comments given by the evaluator, make the necessary modifications and get the approval by the **Evaluator**.
2. If the project topic is rejected, identify a new topic, and request the RP Team for a new topic assessment.